CS-104 Retrieval Quiz (ANSWER KEY)

Fall 2026 · Week 7, Monday

Name: _____

1.

What do the two print statements output?

```
import numpy as np
values = [4, 5]
arr = np.array(values)
print(values * 3)
print(arr * 3)
```

a) `[12, 15]`, then `[12 15]`

b) `[4, 5, 4, 5, 4, 5]`**, then** `[12 15]` **[correct]**

c) `[4, 5, 4, 5, 4, 5]`, then `[4 5 4 5 4 5]`

d) A `TypeError` for both

2.

Which call produces the array `[0. 0.25 0.5 0.75 1.]`?

a) `np.arange(0, 1, 0.25)`

b) `np.linspace(0, 1, 4)`

c) `np.arange(0, 1, 5)`

d) `np.linspace(0, 1, 5)` **[correct]**

3.

What does this code print?

```
import numpy as np
x = np.array([0, 1, 2, 3])
y = x ** 2
print(y)
```

a) `[0 1 4 9]` **[correct]**

b) `[0 2 4 6]`

c) `[0, 1, 2, 3, 0, 1, 2, 3]`

d) It raises a `TypeError`

4.

A radioactive sample decays over time. The amount left after `t` years is

N = 100 × e^(-0.5 t)

Write a code segment that uses NumPy to create 50 evenly spaced times from 0 to 8 years, computes the amount left at each time (use `np.exp`), and uses Matplotlib to plot amount versus time with the axis labels `Time (years)` and `Amount (g)` and a title.

Sample solution:

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 8, 50)
n = 100 * np.exp(-0.5 * t)
plt.plot(t, n)
plt.xlabel("Time (years)")
plt.ylabel("Amount (g)")
plt.title("Radioactive Decay")
plt.show()
```